

Ex.no:4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, conditional filtering, sorting, statistical summary, group aggregation, correlation analysis, and data visualization on a hotel bookings dataset using Pandas and Matplotlib.

Algorithm:

1. Start.
2. Import the required Python libraries: pandas and matplotlib.pyplot.
3. Load the dataset Life Expectancy Data.csv into a pandas DataFrame using `pd.read_csv()`.
4. Inspect the basic structure, initial records, shape, column data types, and numerical summaries using `df.head()`, `df.shape`, `df.dtypes`, and `df.describe()`.
5. Filter rows based on specific conditions, such as countries with Life Expectancy greater than a specified value.
6. Sort the dataset by Life expectancy in descending order using `df.sort_values()`.
7. Compute statistical metrics for Life expectancy including mean, median, mode, and standard deviation.
8. Group data by Status and aggregate to find the average Life expectancy using `df.groupby()`.
9. Calculate the pairwise correlation matrix for numerical features using `df.corr()`.
10. Generate suitable plots for the analysis, such as a bar plot for average Life Expectancy by Status and a scatter plot using Matplotlib.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files

# Upload dataset
uploaded = files.upload()

# Read dataset
df = pd.read_csv("Life Expectancy Data.csv")

# Remove extra spaces from column names
```

```
df.columns = df.columns.str.strip()

print("Dataset loaded successfully!")

# DATA INSPECTION

# Display first five records
print("\nFirst 5 Records:")

display(df.head())

# Display last five records
print("\nLast 5 Records:")

display(df.tail())

# Display dataset shape
print("\nDataset Shape:")

print(df.shape)

print("\nNumber of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

# Display column names
print("\nColumn Names:")

print(df.columns.tolist())

# Display data types
print("\nData Types:")

print(df.dtypes)

# Display dataset information
print("\nDataset Information:")

df.info()

# Display descriptive statistics
print("\nDescriptive Statistics:")

display(df.describe())

# Check missing values
print("\nMissing Values:")

print(df.isnull().sum())

# DATA CLEANING

# Fill missing numerical values with the median
numeric_columns = df.select_dtypes(include=np.number).columns

for column in numeric_columns:

    df[column] = df[column].fillna(df[column].median())

# Fill missing categorical values with the mode
categorical_columns = df.select_dtypes(include="object").columns

for column in categorical_columns:
```

```
if df[column].isnull().sum() > 0:
    df[column] = df[column].fillna(df[column].mode()[0])
print("\nMissing values after cleaning:")
print(df.isnull().sum())

# FILTERING DATA

# Select records where Life Expectancy is greater than 70
high_life_expectancy = df[df["Life expectancy"] > 70]
print("\nRecords with Life Expectancy Greater Than 70:")
display(
    high_life_expectancy[
        ["Country", "Year", "Status", "Life expectancy"]
    ].head(10)
)

# Select records where GDP is greater than 10000
high_gdp = df[df["GDP"] > 10000]
print("\nRecords with GDP Greater Than 10000:")
display(
    high_gdp[
        ["Country", "Year", "GDP", "Life expectancy"]
    ].head(10)
)

# SORTING DATA

# Sort Life Expectancy in descending order
sorted_df = df.sort_values(
    by="Life expectancy",
    ascending=False
)
print("\nLife Expectancy Sorted in Descending Order:")
display(
    sorted_df[
        ["Country", "Year", "Life expectancy"]
    ].head(10)
)

# VISUALIZATION - LIFE EXPECTANCY RANKING

# Select top 10 records
top_10 = sorted_df.head(10)
plt.figure(figsize=(12, 6))
```

```

plt.bar(
    top_10["Country"],
    top_10["Life expectancy"]
)

plt.xlabel("Country")
plt.ylabel("Life Expectancy")
plt.title("Life Expectancy Ranking")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# STATISTICAL ANALYSIS
# Calculate mean
mean_value = df["Life expectancy"].mean()
# Calculate median
median_value = df["Life expectancy"].median()
# Calculate mode
mode_value = df["Life expectancy"].mode()
# Calculate standard deviation
std_value = df["Life expectancy"].std()
print("\nStatistical Analysis of Life Expectancy:")
print("Mean =", round(mean_value, 2))
print("Median =", round(median_value, 2))
print("Mode =", mode_value.tolist())
print("Standard Deviation =", round(std_value, 2))
# Mean Formula:
# Mean = Sum of all values / Number of values
# DESCRIPTIVE STATISTICS
print("\nDescriptive Statistics of Life Expectancy:")
print(df["Life expectancy"].describe())
# GROUPING AND AGGREGATION
# Calculate average Life Expectancy by Status
status_avg = df.groupby(
    "Status"
)["Life expectancy"].mean()
print("\nAverage Life Expectancy by Country Status:")
print(status_avg)

```

```
# VISUALIZATION - STATUS-WISE AVERAGE LIFE EXPECTANCY

plt.figure(figsize=(8, 5))

status_avg.plot(kind="bar")

plt.xlabel("Country Status")

plt.ylabel("Average Life Expectancy")

plt.title("Status-wise Average Life Expectancy")

plt.xticks(rotation=0)

plt.tight_layout()

plt.show()
```

Output:

Saving Life Expectancy Data.csv to Life Expectancy Data (2).csv
Dataset loaded successfully!

First 5 Records:

	Country	Year	Status	Life expectancy	Adult Mortality	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	...	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	thinness 1-19 years	thinness 5-9 years	Income composition of resources	Schooling
0	Afghanistan	2015	Developing	65.0	263.0	62	0.01	71.279624	65.0	1154	...	6.0	8.16	65.0	0.1	584.259210	33736494.0	17.2	17.3	0.479	10.1
1	Afghanistan	2014	Developing	59.9	271.0	64	0.01	73.523582	62.0	492	...	58.0	8.18	62.0	0.1	612.696514	327582.0	17.5	17.5	0.476	10.0
2	Afghanistan	2013	Developing	59.9	268.0	66	0.01	73.219243	64.0	430	...	62.0	8.13	64.0	0.1	631.744976	31731688.0	17.7	17.7	0.470	9.9
3	Afghanistan	2012	Developing	59.5	272.0	69	0.01	78.184215	67.0	2787	...	67.0	8.52	67.0	0.1	669.959000	3696958.0	17.9	18.0	0.463	9.8
4	Afghanistan	2011	Developing	59.2	275.0	71	0.01	7.097109	68.0	3013	...	68.0	7.87	68.0	0.1	63.537231	2978599.0	18.2	18.2	0.454	9.5

5 rows x 22 columns

Last 5 Records:

	Country	Year	Status	Life expectancy	Adult Mortality	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	...	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	thinness 1-19 years	thinness 5-9 years	Income composition of resources	Schooling
2933	Zimbabwe	2004	Developing	44.3	723.0	27	4.36	0.0	68.0	31	...	67.0	7.13	65.0	33.6	454.366654	12777511.0	9.4	9.4	0.407	9.2
2934	Zimbabwe	2003	Developing	44.5	715.0	26	4.06	0.0	7.0	998	...	7.0	6.52	68.0	36.7	453.351155	12633897.0	9.8	9.9	0.418	9.5
2935	Zimbabwe	2002	Developing	44.8	73.0	25	4.43	0.0	73.0	304	...	73.0	6.53	71.0	39.8	57.348340	125525.0	1.2	1.3	0.427	10.0
2936	Zimbabwe	2001	Developing	45.3	686.0	25	1.72	0.0	76.0	529	...	76.0	6.16	75.0	42.1	548.587312	12366165.0	1.6	1.7	0.427	9.8
2937	Zimbabwe	2000	Developing	46.0	665.0	24	1.68	0.0	79.0	1483	...	78.0	7.10	78.0	43.5	547.358878	12222251.0	11.0	11.2	0.434	9.8

Dataset Shape:
(2938, 22)

Number of Rows: 2938
Number of Columns: 22

Column Names:

['Country', 'Year', 'Status', 'Life expectancy', 'Adult Mortality', 'infant deaths', 'Alcohol', 'percentage expenditure', 'Hepatitis B', 'Measles', 'BMI', 'under-five deaths', 'Polio', 'Total expenditure', 'Diphtheria', 'HIV/AIDS', 'GDP', 'Population', 'thinness 1-19 years', 'thinness 5-9 years', 'Income composition of resources', 'Schooling']

Data Types:

```
Country          object
Year             int64
Status           object
Life expectancy  float64
Adult Mortality  float64
infant deaths    int64
Alcohol          float64
percentage expenditure float64
Hepatitis B      float64
Measles          int64
BMI              float64
under-five deaths int64
Polio            float64
Total expenditure float64
Diphtheria       float64
HIV/AIDS         float64
GDP              float64
Population       float64
thinness 1-19 years float64
thinness 5-9 years float64
Income composition of resources float64
Schooling        float64
dtype: object
```

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 2938 entries, 0 to 2937

Data columns (total 22 columns):

#	Column	Non-Null Count	Dtype
0	Country	2938 non-null	object
1	Year	2938 non-null	int64
2	Status	2938 non-null	object
3	Life expectancy	2928 non-null	float64
4	Adult Mortality	2928 non-null	float64
5	infant deaths	2938 non-null	int64
6	Alcohol	2744 non-null	float64
7	percentage expenditure	2938 non-null	float64
8	Hepatitis B	2385 non-null	float64
9	Measles	2938 non-null	int64
10	BMI	2904 non-null	float64
11	under-five deaths	2938 non-null	int64
12	Polio	2919 non-null	float64
13	Total expenditure	2712 non-null	float64
14	Diphtheria	2919 non-null	float64
15	HIV/AIDS	2938 non-null	float64
16	GDP	2490 non-null	float64
17	Population	2286 non-null	float64
18	thinness 1-19 years	2904 non-null	float64
19	thinness 5-9 years	2904 non-null	float64
20	Income composition of resources	2771 non-null	float64
21	Schooling	2775 non-null	float64

dtypes: float64(16), int64(4), object(2)

memory usage: 505.1+ KB

Descriptive Statistics:

	Year	Life expectancy	Adult Mortality	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	BMI	under-five deaths	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population
count	2938.000000	2928.000000	2928.000000	2938.000000	2744.000000	2938.000000	2385.000000	2938.000000	2904.000000	2938.000000	2919.000000	2712.000000	2919.000000	2938.000000	2490.000000	2.286000e+03
mean	2007.518720	69.224932	164.796448	30.303948	4.602861	738.251295	80.940461	2419.592240	38.321247	42.035739	82.550188	5.93819	82.324084	1.742103	7483.158469	1.275338e+07
std	4.613841	9.523867	124.292079	117.926501	4.052413	1987.914858	25.070016	11467.272489	20.044034	160.445548	23.428046	2.49832	23.716912	5.077785	14270.169342	6.101210e+07
min	2000.000000	36.300000	1.000000	0.000000	0.010000	0.000000	1.000000	0.000000	1.000000	0.000000	3.000000	0.37000	2.000000	0.100000	1.681350	3.400000e+01
25%	2004.000000	63.100000	74.000000	0.000000	0.877500	4.685343	77.000000	0.000000	19.300000	0.000000	78.000000	4.26000	78.000000	0.100000	463.935626	1.957932e+05
50%	2008.000000	72.100000	144.000000	3.000000	3.755000	64.912906	92.000000	17.000000	43.500000	4.000000	93.000000	5.75500	93.000000	0.100000	1766.947595	1.386542e+06
75%	2012.000000	75.700000	228.000000	22.000000	7.702500	441.534144	97.000000	360.250000	56.200000	28.000000	97.000000	7.49250	97.000000	0.800000	5910.806335	7.420359e+06
max	2015.000000	89.000000	723.000000	1800.000000	17.870000	19479.911610	99.000000	212183.000000	87.300000	2500.000000	99.000000	17.60000	99.000000	50.600000	119172.741800	1.293859e+09

Records with Life Expectancy Greater Than 70:

	Country	Year	Status	Life expectancy
16	Albania	2015	Developing	77.8
17	Albania	2014	Developing	77.5
18	Albania	2013	Developing	77.2
19	Albania	2012	Developing	76.9
20	Albania	2011	Developing	76.6
21	Albania	2010	Developing	76.2
22	Albania	2009	Developing	76.1
23	Albania	2008	Developing	75.3
24	Albania	2007	Developing	75.9
25	Albania	2006	Developing	74.2

Records with GDP Greater Than 10000:

	Country	Year	GDP	Life expectancy
64	Antigua and Barbuda	2015	13566.95410	76.4
65	Antigua and Barbuda	2014	12888.29667	76.2
66	Antigua and Barbuda	2013	12224.86416	76.1
67	Antigua and Barbuda	2012	12565.44197	75.9
68	Antigua and Barbuda	2011	11929.34991	75.7
69	Antigua and Barbuda	2010	12126.87614	75.6
72	Antigua and Barbuda	2007	14252.29452	75.0
73	Antigua and Barbuda	2006	12724.39377	74.8
74	Antigua and Barbuda	2005	11371.93895	74.6
80	Argentina	2015	13467.12360	76.3

Life Expectancy Sorted in Descending Order:

	Country	Year	Life expectancy
2056	Portugal	2014	89.0
995	Germany	2014	89.0
2433	Spain	2007	89.0
1850	New Zealand	2010	89.0
915	Finland	2014	89.0
241	Belgium	2014	89.0
937	France	2008	89.0
938	France	2007	89.0
2513	Sweden	2007	89.0
1293	Italy	2004	89.0

Statistical Analysis of Life Expectancy:

Mean = 69.23
Median = 72.1
Mode = [73.0]
Standard Deviation = 9.51

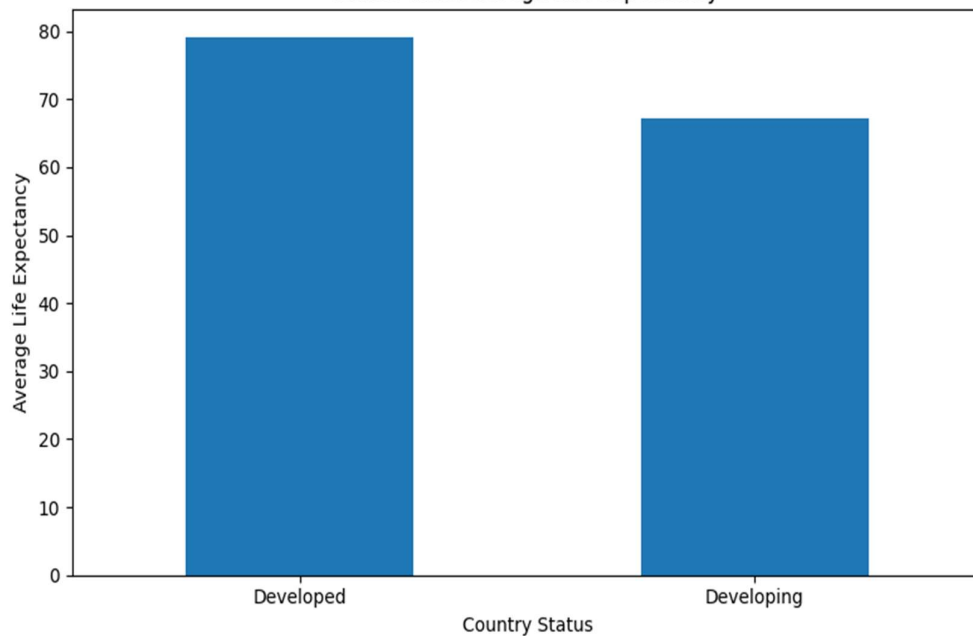
Descriptive Statistics of Life Expectancy:

count 2938.000000
mean 69.234717
std 9.509115
min 36.300000
25% 63.200000
50% 72.100000
75% 75.600000
max 89.000000
Name: Life expectancy, dtype: float64

Average Life Expectancy by Country Status:

Status
Developed 79.197852
Developing 67.132028
Name: Life expectancy, dtype: float64

Status-wise Average Life Expectancy



Top 10 Countries by Average Life Expectancy:

Country	Average Life Expectancy
Japan	82.53750
Sweden	82.51875
Iceland	82.44375
Switzerland	82.33125
France	82.21875
Italy	82.18750
Spain	82.06875
Australia	81.81250
Norway	81.79375
Canada	81.68750

Name: Life expectancy, dtype: float64

Correlation with Life Expectancy:

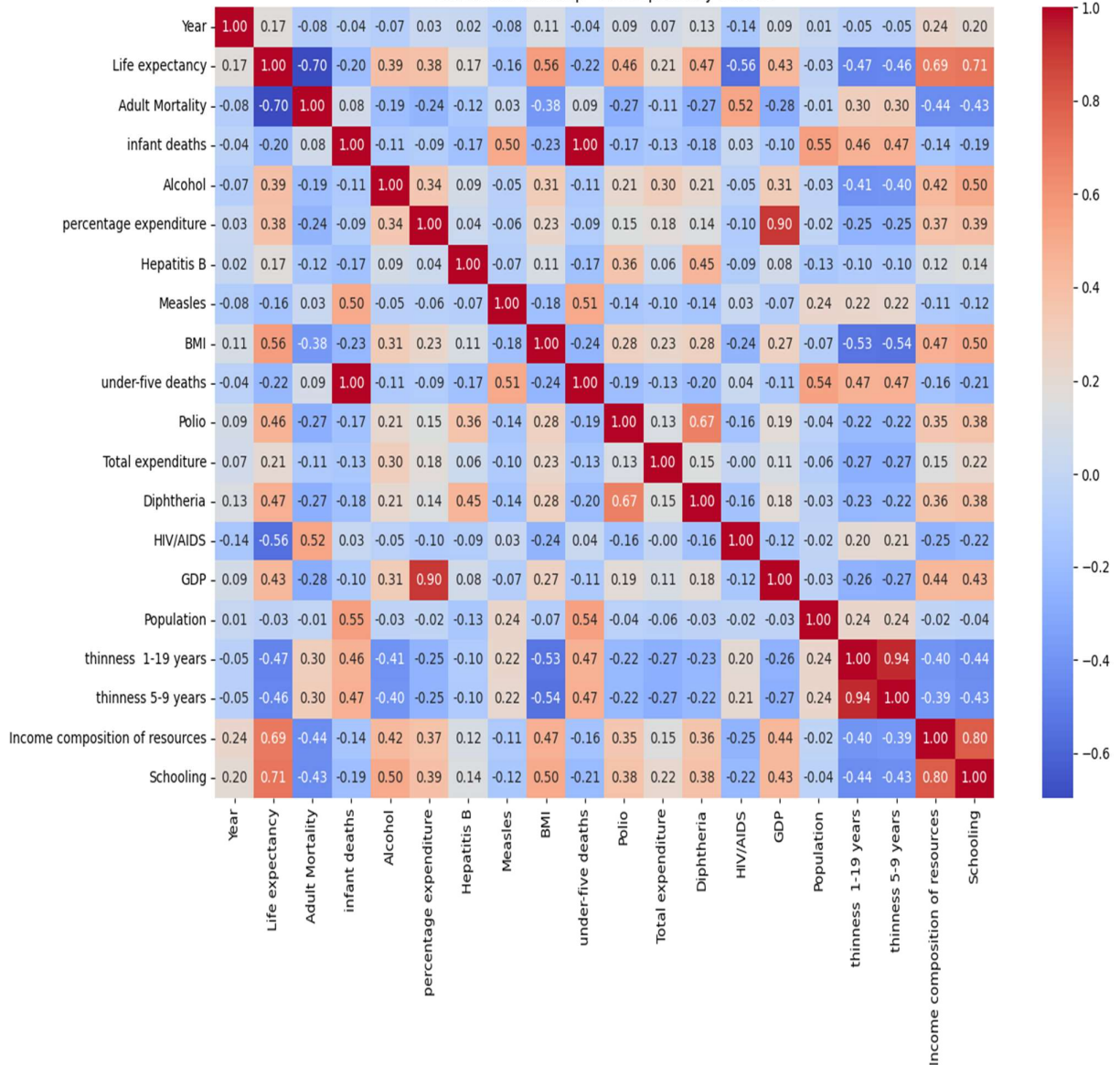
Life expectancy	1.000000
Schooling	0.713054
Income composition of resources	0.688662
BMI	0.556901
Diphtheria	0.472211
Polio	0.458399
GDP	0.430461
Alcohol	0.388918
percentage expenditure	0.381418
Total expenditure	0.208844
Year	0.170819
Hepatitis B	0.170219
Population	-0.029014
Measles	-0.157767
infant deaths	-0.196769
under-five deaths	-0.222738
thinness 5-9 years	-0.462473
thinness 1-19 years	-0.468002
HIV/AIDS	-0.556703
Adult Mortality	-0.696390

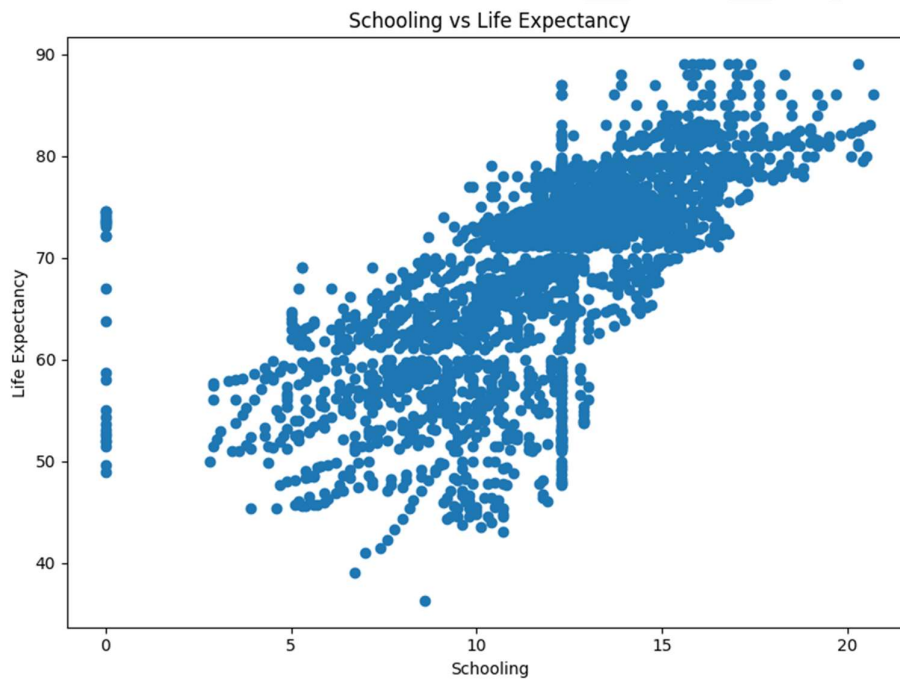
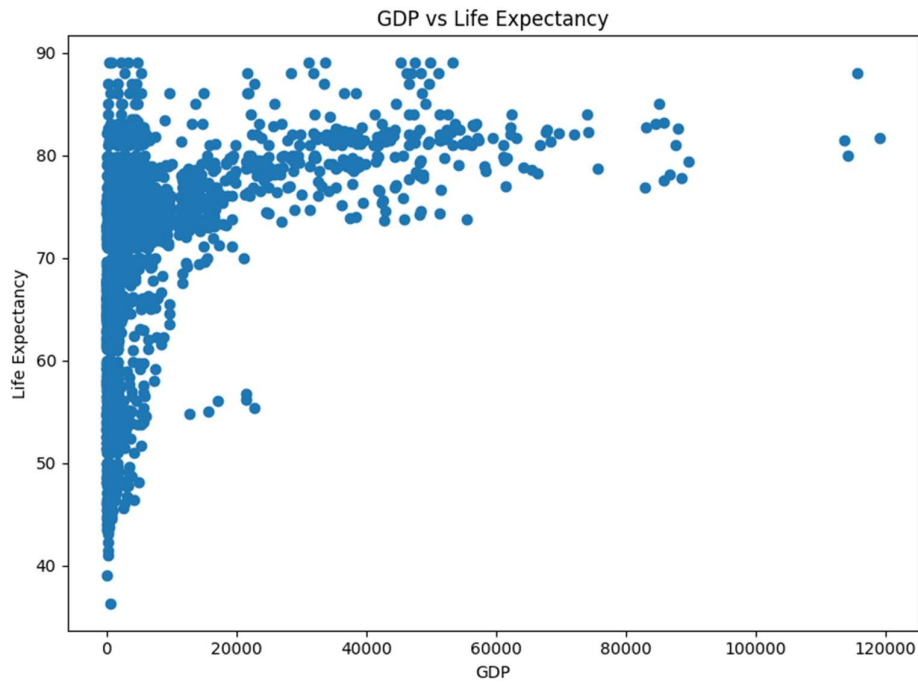
Name: Life expectancy, dtype: float64

Correlation Matrix:

	Year	Life expectancy	Adult Mortality	infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	BMI	under-five deaths	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	thinness 1-19 years	thinness 5-9 years	Income composition of resources	Schooling
Year	1.00000	0.17019	-0.07933	-0.07415	-0.069147	0.031400	0.021302	-0.00493	0.108484	-0.042937	0.091388	0.074552	0.131373	-0.139741	0.091480	0.014027	-0.047827	-0.050859	0.236594	0.203517
Life expectancy	0.17019	1.00000	-0.69390	-0.196769	0.388918	0.381418	0.170219	-0.157787	0.556901	-0.222738	0.458399	0.208844	0.472211	-0.556703	0.430461	-0.029014	-0.468002	-0.462473	0.688662	0.713054
Adult Mortality	-0.07933	-0.69390	1.00000	0.078890	-0.190191	-0.242628	-0.123287	0.031293	-0.379092	0.094280	-0.270002	-0.111796	-0.270301	0.523886	-0.281434	-0.005292	0.296135	0.301634	-0.436332	-0.433415
infant deaths	-0.07415	-0.196769	0.078890	1.00000	-0.112918	-0.085612	-0.167421	0.501128	-0.228796	0.996629	-0.170896	-0.126268	-0.175376	0.025231	-0.102829	0.551667	0.464811	0.470445	-0.141571	-0.190951
Alcohol	-0.069147	0.388918	-0.190191	-0.112918	1.00000	0.344410	0.090850	-0.049752	0.313022	-0.109750	0.211744	0.296153	0.212388	-0.046235	0.313060	-0.026836	-0.413021	-0.402034	0.418372	0.498099
percentage expenditure	0.031400	0.381418	-0.242628	-0.085612	0.344410	1.00000	0.039528	-0.056596	0.227408	-0.087852	0.146148	0.175559	0.142506	-0.097857	0.901819	-0.016888	-0.249610	-0.251144	0.374730	0.386076
Hepatitis B	0.021302	0.170219	-0.123287	-0.167421	0.090850	0.039528	1.00000	-0.073021	0.112441	-0.170549	0.364890	0.055988	0.449151	-0.085497	0.076660	-0.125006	-0.097576	-0.100576	0.117652	0.141275
Measles	-0.00493	-0.157787	0.031293	0.501128	-0.049752	-0.056596	-0.073021	1.00000	-0.176069	0.507809	-0.136669	-0.104133	-0.142385	0.030899	-0.069253	0.237188	0.224700	0.221138	-0.111089	-0.120799
BMI	0.108484	0.556901	-0.379092	-0.228796	0.313022	0.227408	0.112441	-0.176069	1.00000	-0.237048	0.281260	0.230745	0.280267	-0.243383	0.273932	-0.069667	-0.532487	-0.539365	0.471947	0.499806
under-five deaths	-0.042937	-0.222738	0.094280	0.996629	-0.109750	-0.087852	-0.170549	0.507809	-0.237048	1.00000	-0.188864	-0.127960	-0.195808	0.038062	-0.106093	0.539260	0.466697	0.471167	-0.159264	-0.206242
Polio	0.091388	0.458399	-0.270002	-0.170896	0.211744	0.146148	0.364890	-0.136669	0.281260	-0.188864	1.00000	0.131037	0.673977	-0.159128	0.189926	-0.044813	-0.218401	-0.219168	0.345172	0.375830
Total expenditure	0.074552	0.208844	-0.111796	-0.126268	0.296153	0.175559	0.055988	-0.104133	0.230745	-0.127960	0.131037	1.00000	0.146485	-0.000289	0.112099	-0.061602	-0.267572	-0.274069	0.154328	0.220170
Diphtheria	0.131373	0.472211	-0.270301	-0.175376	0.212388	0.142506	0.449151	-0.142385	0.280267	-0.195808	0.673977	0.146485	1.00000	-0.164419	0.181953	-0.033128	-0.226409	-0.219699	0.359379	0.379062
HIV/AIDS	-0.139741	-0.556703	0.523886	0.025231	-0.046235	-0.097857	-0.085497	0.030899	-0.243383	0.038062	-0.159128	-0.000289	-0.164419	1.00000	-0.122590	-0.017094	0.203684	0.206899	-0.247823	-0.218822
GDP	0.091480	0.430461	-0.005292	-0.102829	0.313060	0.901819	0.076660	-0.069253	0.273932	-0.106093	0.189926	0.112099	0.181953	-0.122590	1.00000	-0.025269	-0.263323	-0.266935	0.435960	0.432229
Population	0.014027	-0.029014	-0.005292	0.551667	-0.026836	-0.016888	-0.125006	0.237188	-0.069667	0.539260	-0.044813	-0.061602	-0.033128	-0.017094	-0.025269	1.00000	0.236410	0.235278	-0.017237	-0.036814
thinness 1-19 years	-0.047827	-0.468002	0.296135	0.464811	-0.413021	-0.249610	-0.097576	0.224700	-0.532487	0.466697	-0.218401	-0.267572	-0.226409	0.203684	-0.263323	0.236410	1.00000	0.939187	-0.401427	-0.427126
thinness 5-9 years	-0.050859	-0.462473	0.301634	0.470445	-0.402034	-0.251144	-0.100576	0.221138	-0.539365	0.471167	-0.219168	-0.274069	-0.219699	0.206899	-0.266935	0.235278	0.939187	1.00000	-0.390961	-0.426925
Income composition of resources	0.236594	0.688662	-0.436332	-0.141571	0.418372	0.374730	0.117652	-0.110899	0.471947	-0.159264	0.345172	0.154328	0.359379	-0.247823	0.435960	-0.017237	-0.401437	-0.390961	1.00000	0.795383
Schooling	0.203517	0.713054	-0.433415	-0.190951	0.498099	0.386076	0.141275	-0.120799	0.499806	-0.206242	0.375830	0.220170	0.379062	-0.218822	0.432229	-0.036814	-0.437126	-0.426925	0.795383	1.00000

Correlation Heatmap - Life Expectancy Dataset





Dataset Shape: (2938, 22)

Total Records: 2938

Total Features: 22

Mean Life Expectancy: 69.23
Median Life Expectancy: 72.1
Standard Deviation: 9.51

Result: